Question	Answer	Marks	Partial Marks
8(b)	154 or 154.3 to 154.4	**3**	**M2** for $450 \times \left(\sqrt{\frac{98}{200}}\right)^3$ oe or **M1** for $\left(\sqrt{\frac{98}{200}}\right)^3$ or $\left(\sqrt{\frac{200}{98}}\right)^3$ oe or for $\left(\frac{450}{V}\right)^2 = \left(\frac{200}{98}\right)^3$ oe
9(a)(i)	0	**1**	
9(a)(ii)	$\frac{1}{7}$ oe	**1**	
9(b)	$\frac{4}{49}$ oe	**2**	**M1** for $\frac{2}{7} \times \frac{2}{7}$
9(c)(i)	$\frac{2}{21}$ oe	**3**	**M2** for $\frac{2}{7} \times \frac{1}{6} + \frac{1}{7} \times \frac{2}{6}$ oe or **M1** for $\frac{2}{7} \times \frac{1}{6}$ or $\frac{1}{7} \times \frac{2}{6}$ oe seen If 0 scored SC1 for $\frac{4}{49}$
9(c)(ii)	$\frac{19}{21}$ oe	**3**	**M2** for $1 - \left(\frac{2}{7} \times \frac{1}{6}\right) - \left(\frac{2}{7} \times \frac{1}{6}\right)$ oe or **M1** for $\left(\frac{2}{7} \times \frac{1}{6}\right) + \left(\frac{2}{7} \times \frac{1}{6}\right)$ oe ALTERNATIVE **M2** for $\frac{1}{7}$ [× 1] × 3 + $\frac{2}{7} \times \frac{5}{6}$ × 2 or **M1** for $\frac{2}{7} \times \frac{5}{6}$ or $\frac{1}{7}$ [× 1] × 3 If 0 scored SC1 for $\frac{38}{49}$
9(d)	3	**2**	**M1** for $\frac{5}{7} \times \frac{4}{6} \times \frac{2\text{or}3}{5}$
10(a)	$7x^6 - 42x^5$ final answer	**2**	**B1** for one correct term $7x^6$ or $42x^5$ or for $7x^6 - 42x^5$ seen and spoiled

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Question	Answer	Marks	Partial Marks
10(b)	$49x + 41$	**4**	**M1** for substituting $x = -1$ into $[y =]\ x^7 - 7x^6$ **M1** for $x = -1$ substituted in *their* (a) or the correct derivative to give *their m* **M1** for *their* $-8 = (\textit{their } m)(-1) + c$ oe
10(c)	(0, 0) (6, –46 656)	**5**	**B4** for (6, –46 656) or **B3** for $x = 0$ and 6 **OR** **M1** for *their* $\frac{dy}{dx} = 0$ or stating $\frac{dy}{dx} = 0$ and **M1** for a correct method to solve *their* $7x^6 - 42x^5$
11(a)	$\frac{3}{10} \times 360$ oe	**M2**	**M1** for $\frac{3}{3+7} = \frac{x}{360}$ or for $\frac{x}{360}\ [\times 2\pi r] = \frac{3}{7} \times \frac{360-x}{360}\ [\times 2\pi r]$ oe or better or $1\ [\times 2\pi r] = \frac{10}{7} \times \frac{360-x}{360}\ [\times 2\pi r]$ oe or better or $\frac{360}{7+3} \times k$ ($k = 1$ or 7)
	108	**A1**	
11(b)(i)	$\frac{1}{2}r^2 \sin y = \frac{1}{2} \times \frac{y}{360} \times \pi r^2$ or $\frac{y}{360} \times \pi r^2 = [2 \times \frac{1}{2}]\ r^2 \sin y$ and one further step leading to $360\sin y = \pi y$ with no errors	**2**	**M1** for $\frac{y}{360} \times \pi r^2$ or for $\frac{1}{2}r^2 \sin y$
11(b)(ii)	341.18 or 341.22 341.00 341.49 or 341.54	**3**	**B1** for each
11(b)(iii)	108.6 cao	**1**	